

DESKTOP PUBLISHING AND DESIGNING

Unit-IV: Vector Images, Color Modes, Photoshop Tools & Layers

Comprehensive Study Notes

HPU BCA Semester III

UNIT-IV: VECTOR IMAGES, COLOR MODES, PHOTOSHOP TOOLS & LAYERS

1. Raster and Vector Images

1.1 What are Raster Images?

Raster Images (also called **Bitmap Images**) are made up of a grid of tiny colored squares called **pixels**. Each pixel contains color information, and the image is stored as a collection of pixels.

Characteristics of Raster Images

Characteristic	Description
Made of Pixels	Composed of individual colored squares
Resolution Dependent	Quality depends on DPI/PPI
Loss of Quality	Distorts when enlarged
File Size	Large file sizes
Editing	Pixel-level editing possible

Common Raster Formats

Format	Extension	Best Use
JPEG	.jpg, .jpeg	Photos, web images
PNG	.png	Web graphics, transparent images
GIF	.gif	Animations, simple graphics
BMP	.bmp	Windows images

TIFF	.tif, .tiff	High-quality print images
PSD	.psd	Photoshop editing

1.2 What are Vector Images?

Vector Images are made up of mathematical paths, points, lines, curves, and shapes. They are resolution-independent and can be scaled infinitely without losing quality.

Characteristics of Vector Images

Characteristic	Description
Mathematical Paths	Composed of lines and curves
Resolution Independent	Can scale infinitely
No Quality Loss	Maintains quality at any size
Small File Size	Smaller than raster images
Editing	Individual elements can be edited

Common Vector Formats

Format	Extension	Best Use
EPS	.eps	Print design
SVG	.svg	Web design, scalable
AI	.ai	Adobe Illustrator
CDR	.cdr	CorelDRAW
WMF	.wmf	Windows metafile
PDF	.pdf	Documents with vector elements

1.3 Raster vs Vector Comparison

Feature	Raster Image	Vector Image
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Composition	Pixels	Mathematical paths
Scalability	Loses quality when enlarged	Infinite scaling
File Size	Large	Small
Editing	Pixel-level editing	Object-level editing
Best For	Photos, detailed images	Logos, illustrations, text
Software	Photoshop, GIMP	Illustrator, CorelDRAW
Formats	JPEG, PNG, GIF, BMP	EPS, SVG, AI, CDR

2. Color Modes

2.1 What are Color Modes?

Color Modes are systems for representing colors in digital images. They define how colors are created and displayed.

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+-----+          +-----+
? RGB MODE      ?    ? CMYK MODE    ?
? (Screen/Web) ?    ? (Print)      ?
+-----+          +-----+
+-----+          +-----+
? GRAYSCALE MODE?    ? LAB MODE      ?
? (Black & White)?    ? (Device      ?
?                   ?    ? Independent)?
+-----+          +-----+

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2.2 RGB Color Model

RGB (Red, Green, Blue) is the color model used for digital displays, monitors, TVs, and web design. It is an additive color model where colors are created by adding light.

RGB Color Values

Color	Red	Green	Blue
Black	0	0	0
White	255	255	255
Red	255	0	0
Green	0	255	0
Blue	0	0	255
Yellow	255	255	0

Cyan	0	255	255
Magenta	255	0	255
Gray	128	128	128

2.3 CMYK Color Model

CMYK (Cyan, Magenta, Yellow, Black) is the color model used for printing. It is a subtractive color model where colors are created by subtracting light using ink.

CMYK Color Values

Color	Cyan	Magenta	Yellow	Black
Black	0	0	0	100
White	0	0	0	0
Red	0	100	100	0
Green	100	0	100	0
Blue	100	100	0	0
Yellow	0	0	100	0
Cyan	100	0	0	0
Magenta	0	100	0	0

2.4 RGB vs CMYK Comparison

Feature	RGB	CMYK
Color Model	Additive	Subtractive
Primary Colors	Red, Green, Blue	Cyan, Magenta, Yellow, Black
Use	Screen/Web	Print
Color Range	Wider (more colors)	Narrower
File Size	Smaller	Larger
Software	Photoshop, Web	Photoshop, Print Software

2.5 Other Color Modes

- **Grayscale Mode:** Uses shades of gray from black to white. Used for black-and-white images.
- **Indexed Color Mode:** Uses a limited color palette (up to 256 colors). Produces smaller file sizes for web graphics.
- **Lab Color Mode:** Device-independent color mode based on human vision. Consists of L (lightness), a (green-red), and b (blue-yellow) channels.
- **Bitmap Mode:** Uses only black and white pixels. Used for line art and text.
- **Duotone Mode:** Uses two to four custom inks to create enhanced grayscale images.

3. Photoshop Interface

3.1 Photoshop Screen Components

Component	Description
Application Bar	Menu bar, document name, zoom, screen mode
Options Bar	Tool-specific settings and options
Toolbox	Contains all Photoshop tools
Canvas/Work Area	Main area for editing images
Layers Panel	Manage layers of the document
Colors Panel	Select foreground and background colors
History Panel	Undo/redo history
Status Bar	Document information, zoom level

3.2 Photoshop Screen Modes

Mode	Description
Standard Screen Mode	Default mode with menus and panels
Full Screen Mode with Menu Bar	Full screen with menu bar
Full Screen Mode	Complete full screen without distractions

4. Photoshop Tools

4.1 Painting Tools

Tool	Shortcut	Function
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Brush	B	Paint strokes with soft or hard edges
Pencil	B	Paint hard-edged strokes
Color Replacement	B	Replace specific colors in an image

4.2 Editing Tools

Tool	Shortcut	Function
Healing Brush	J	Repair imperfections by blending pixels
Patch	J	Repair larger areas by replacing with sampled pixels
Clone Stamp	S	Duplicate exact pixels from one area to another

4.3 Retouching Tools

Tool	Shortcut	Function
Blur	R	Soften areas by reducing detail
Sharpen	R	Enhance detail by increasing contrast
Smudge	R	Smear pixels like dragging through wet paint
Dodge	O	Lighten areas of the image
Burn	O	Darken areas of the image
Sponge	O	Saturate or desaturate colors

5. Layers

5.1 What are Layers?

Layers are like transparent sheets stacked on top of each other. Each layer can contain different elements that can be edited independently. Layers are fundamental to non-destructive editing in Photoshop.

5.2 Types of Layers

Layer Type	Description
Background Layer	The base layer; usually locked
Text Layer	Contains editable text
Adjustment Layer	Applies color/tonal adjustments non-destructively
Fill Layer	Filled with solid color, gradient, or pattern
Shape Layer	Contains vector shapes
Smart Object	Preserves original image data for non-destructive editing

5.3 Layer Properties

Property	Description
Layer Name	Name of the layer
Visibility	Toggle visibility (eye icon)
Lock	Lock layer (prevents editing)
Opacity	Transparency (0% to 100%)
Fill	Fill opacity
Blend Mode	How layer interacts with layers below

Layer Style	Effects (drop shadow, stroke, etc.)
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5.4 Layer Operations

Operation	How To Do
Create Layer	Layers → New → Layer
Delete Layer	Right-click → Delete Layer
Duplicate Layer	Right-click → Duplicate Layer
Merge Layers	Select layers → Merge
Layer Order	Drag layer up/down
Group Layers	Select layers → Group
Link Layers	Select layers → Link
Create Adjustment Layer	Layers → New Adjustment Layer

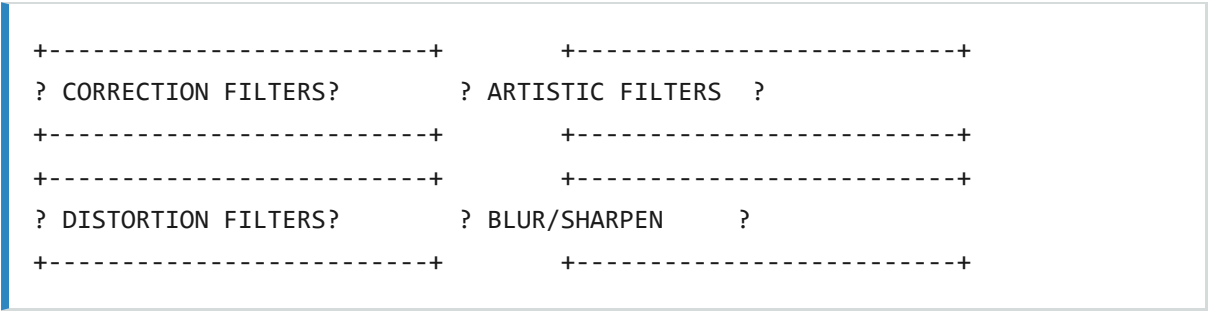
5.5 Layer Blend Modes

Blend Mode	Effect
Normal	Default mode
Multiply	Darkens colors
Screen	Lightens colors
Overlay	Combines multiply and screen
Soft Light	Soft contrast
Hard Light	Hard contrast
Difference	Shows difference between colors
Hue	Applies hue only

6. Filters

6.1 What are Filters?

Filters are special effects that can be applied to images to create various artistic and correction effects.



6.2 Filter Types

Filter Category	Examples	Use
Artistic	Watercolor, Oil Paint	Artistic effects
Blur	Gaussian Blur, Motion Blur	Soften images
Brush Strokes	Crosshatch, Spatter	Brush stroke effects
Distort	Twirl, Ripple, Wave	Distortion effects
Noise	Add Noise, Despeckle	Noise effects
Pixelate	Mosaic, Pointillize	Pixelation effects
Render	Lens Flare, Clouds, Lighting	Render effects
Sharpen	Sharpen, Unsharp Mask	Enhance details
Sketch	Graphic Pen, Note Paper	Sketch effects
Stylize	Emboss, Solarize, Edge Detect	Stylized effects
Texture	Grain, Mosaic, Patchwork	Texture effects

7. Image Editing Workflow & Operations

7.1 Common Image Operations

Operation	How To
Resize	Image → Image Size
Crop	Crop tool
Adjust Brightness	Image → Adjustments → Brightness/Contrast
Adjust Color	Image → Adjustments → Color Balance
Resize Canvas	Image → Canvas Size
Rotate	Image → Rotate Canvas
Layer Mask	Layers → Layer Mask
Transform	Edit → Transform → Scale/Rotate/Skew

7.2 Image Adjustments

Adjustment	Use
Brightness/Contrast	Basic tonal adjustment
Levels	Advanced tonal adjustment
Curves	Precise tonal control
Color Balance	Adjust color casts
Hue/Saturation	Adjust color intensity
Desaturate	Remove color
Invert	Invert colors

Threshold	Convert to black and white
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8. File Formats in DTP

Common file formats used in Desktop Publishing include:

Format	Type	Use in DTP
JPEG	Raster	Photographs and images in publications
PNG	Raster	Web graphics and images with transparency
TIFF	Raster	High-quality print images
EPS	Vector	Logos and illustrations in print
PDF	Both	Final document output for print and sharing
AI	Vector	Illustrator graphics for import
PSD	Raster	Photoshop source files for editing

Questions

Short Answer Questions (2-5 Marks)

1. What is the difference between raster and vector images?
2. What is RGB color model?
3. What is CMYK color model?
4. What are layers in Photoshop?
5. What are filters in Photoshop?
6. What is the Toolbox in Photoshop?
7. What are Photoshop palettes?
8. What is the difference between RGB and CMYK?
9. What is a layer mask?
10. What are blend modes in Photoshop?

Long Answer Questions (10 Marks)

1. Explain raster and vector images with their differences.
2. Explain RGB and CMYK color modes with their uses.
3. Explain the Photoshop toolbox with different tools.
4. Explain layers and their operations in Photoshop.
5. Explain filters and their categories in Photoshop.
6. Explain the Photoshop workspace and its components.

Objective Questions (1 Mark)

1. RGB stands for:
(a) Red, Green, Blue (b) Red, Gray, Black (c) Red, Green, Black (d) None
2. CMYK stands for:
(a) Cyan, Magenta, Yellow, Black (b) Cyan, Magenta, Yellow, Blue (c) Cyan, Magenta, Yellow, Green (d) None

3. PNG is a:

(a) Vector format (b) Raster format (c) Both (d) None

4. Which tool is used to move layers?

(a) Brush (b) Move (c) Lasso (d) Crop

5. SVG is a:

(a) Vector format (b) Raster format (c) Both (d) None

Quick Revision

Raster vs Vector

Feature	Raster	Vector
Made of	Pixels	Paths
Scalability	Limited	Infinite
File Size	Large	Small
Best For	Photos	Logos

Color Modes

Color Modes:

- RGB: Red, Green, Blue (Additive - Screen/Web)
- CMYK: Cyan, Magenta, Yellow, Black (Subtractive - Print)
- Grayscale: Shades of gray
- Lab: Device-independent
- Indexed: 256 colors max
- Bitmap: Black and white only
- Duotone: Two to four custom inks

Photoshop Key Tools

Selection: Marquee (M), Lasso (L), Magic Wand (W), Quick Selection

Painting: Brush (B), Pencil (B), Color Replacement

Editing: Healing Brush (J), Patch (J), Clone Stamp (S)

Retouching: Blur (R), Sharpen (R), Smudge (R), Dodge (O), Burn (O), Sponge (O)

Other: Crop (C), Eyedropper (I), Gradient (G), Pen (P), Type (T), Shape (U)

Layer Operations

Types: Background, Text, Adjustment, Fill, Shape, Smart Object

Operations: Create, Delete, Duplicate, Merge, Group, Link, Blend Modes

Properties: Name, Visibility, Lock, Opacity, Fill, Layer Style

End of Unit-IV Notes